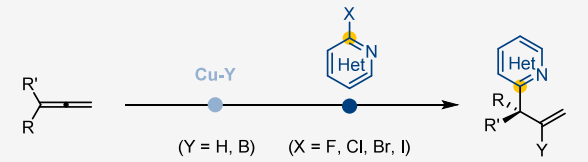
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Copper-Catalyzed Enantioselective Synthesis of *N*-Heterocycle-Substituted Quaternary Carbon Stereogenic Centers

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ABSTRACT: The enantioselective construction of all-carbon quaternary stereogenic centers adjacent to *N*-heterocycles remains a long-standing challenge in asymmetric catalysis. Herein we report a copper-catalyzed enantioselective hydropyridylation of 1,1-disubstituted allenes with 2-halopyridines that proceeds via a nucleophilic aromatic substitution (S_NAr) pathway. This transformation enables the efficient synthesis of α-quaternary *N*-heterocyclic compounds with a broad substrate scope in high yields (up to 96%) with excellent enantioselectivities (up to 99% *ee*). The methodology is compatible with various functional groups and *N*-heterocycles and can be extended to copper-catalyzed boropyridylation, affording versatile alkenylboronates. Mechanistic studies, including deuterium labeling, kinetic isotope effect analysis, and DFT calculations, support a 2,3-hydrocupration followed by a stereodetermining S_NAr step. This strategy provides a practical and general approach for constructing structurally diverse azaarene-containing quaternary centers, which are potentially valuable for drug discovery and development.



■ Quaternary stereogenic center ■ 49 examples, up to 96% yield, 99% ee

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■ INTRODUCTION

N-Heterocyclic compounds in which an *N*-heterocycle is attached to a quaternary stereogenic carbon center represent privileged structural motifs that are widely distributed in biologically active natural products and serve as key pharmacophores in medicinal chemistry (Figure 1a).[1] The construction of such architecturally complex motifs is of considerable synthetic importance, as the introduction of a quaternary carbon stereocenter adjacent to a nitrogen-containing heterocycle imparts significant conformational rigidity and can markedly influence the biological and physicochemical properties of the resulting molecules. Over the past few decades, substantial efforts have been devoted to the development of efficient and stereoselective methods for accessing these challenging stereocenters.

Among the established strategies, stereospecific cross-coupling reactions using enantioenriched substrates[2−5] as well as α-functionalization of prochiral compounds containing azaarene motifs[6,7] have emerged as effective tools. However, these methodologies often exhibit limitations with respect to substrate scope, particularly in accommodating diverse heterocyclic frameworks, and tend to require multistep preparation of the starting materials or stoichiometric chiral auxiliaries (Figure 1b, left). In this context, asymmetric transition metal catalysis has provided a robust platform to address the inherent challenges associated with the formation of all-carbon quaternary stereocenters adjacent to nitrogen atoms. In particular, nickel-catalyzed asymmetric heteroarylation has been reported as a promising strategy for the stereoselective α-functionalization of azaarenes.[8] Despite its effectiveness, this approach typically requires high reaction temperatures and is predominantly limited to substrates with preassembled cyclic backbones, thereby restricting structural diversification (Figure 1b, right).

Alternatively, copper catalysis has emerged as a sustainable and cost-effective approach to asymmetric catalysis. Recent advances have showcased the potential of copper-catalyzed transformations involving π-unsaturated systems—such as alkenes, dienes, and allenes—for the construction of quaternary carbon stereocenters.[9] In our previous work, we explored copper-catalyzed diene functionalization;[10] however, steric congestion around the reaction site impeded the formation of quaternary centers.[11] Allenes, by contrast, have shown considerable promise in copper-catalyzed coupling reactions with carbonyl and imine electrophiles, enabling the formation of fully substituted carbon centers under mild conditions.[9c,12] Nevertheless, despite their synthetic potential, the application of allene-based methodologies to azaarene-derived substrates remains scarcely explored, with only a single report to date describing a limited example involving indazole derivatives[13] (Figure 1c).

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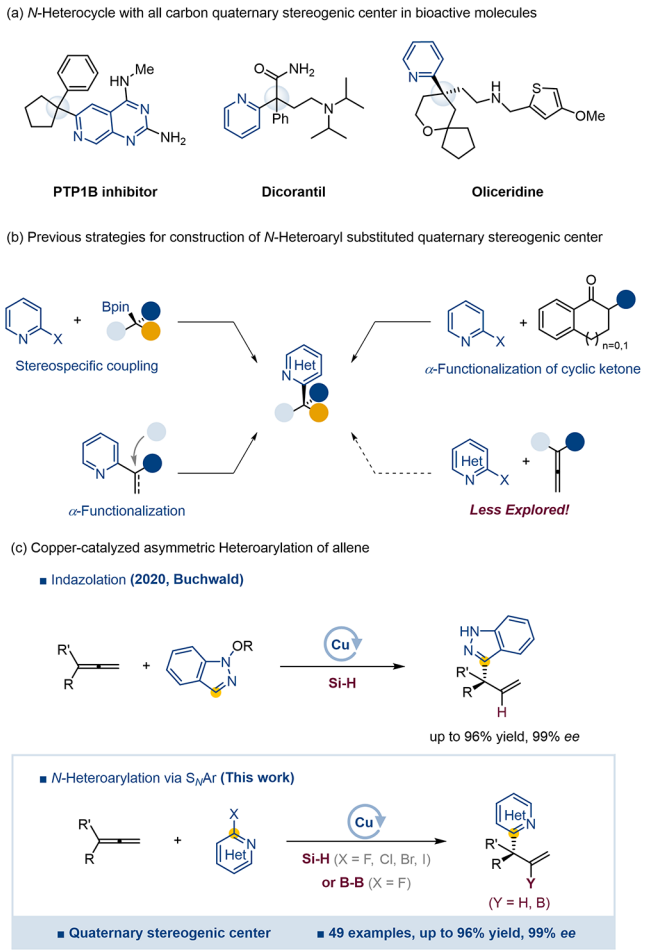


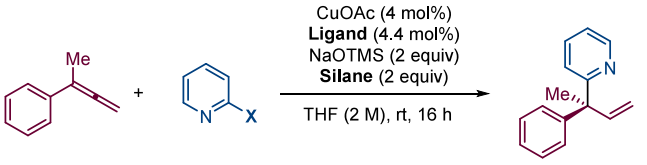
Figure 1. Previous work on the construction of quaternary stereogenic centers containing an *N*-heterocycle.

Motivated by these findings, we envisioned a new approach that would enable the stereoselective construction of α-quaternary *N*-heterocycles through copper-catalyzed allene functionalization. Central to our design is the innate coordinating ability of the nitrogen atom within the azaarene ring, which we anticipated could serve to both activate the substrate and direct the catalytic transformation with high regio- and stereocontrol. Herein we report the development of a copper-catalyzed stereoselective synthesis of *N*-heterocyclic compounds bearing an all-carbon quaternary stereogenic center through the regio- and stereocontrolled functionalization of allenes. This transformation not only expands the synthetic utility of copper catalysis in constructing highly congested stereocenters but also offers a general and efficient route to structurally diverse azaarene derivatives that may be of value in drug discovery and related fields.

RESULTS AND DISCUSSION

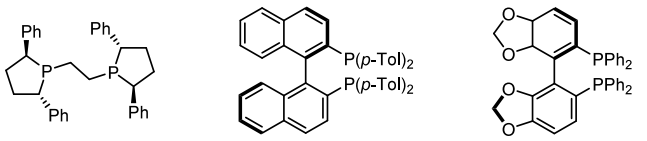
We initiated our investigation of copper-catalyzed hydropyridylation using 1-phenyl-1-methylallene (**1a**) and 2-fluoropyridine (**2a**) in the presence of CuOAc (4 mol %), (*S*,*S*)-Ph-BPE (4.4 mol %), NaOTMS (2.0 equiv), and Ph_3SiH (2.0 equiv) in THF at room temperature for 16 h. Screening a series of chiral phosphine ligands—including (*S*,*S*)-Ph-BPE, (*R*)-*p*-Tol-BINAP, and (*R*)-SEGPHOS—yielded the desired product **3a** in 76%, 31%, and 68% yield with 94%, 64%, and 44% enantiomeric excess (*ee*), respectively (Table 1, entries 1–3). Among these, (*S*,*S*)-Ph-BPE afforded the highest yield and enantioselectivity. Replacing Ph_3SiH with Ph_2MeSiH slightly improved the yield to 77%, whereas use of $(EtO)_2MeSiH$ resulted in a markedly diminished yield (33%) and reduced enantioselectivity (88% *ee*), accompanied by the formation of 2-ethoxypyridine as a side product (Table 1, entries 4-6; see Table S2). Notably, switching the solvent to 1,4-dioxane led to an increased yield of 86% (Table 1, entry 7). To evaluate the influence of the leaving group on the pyridine electrophile, various 2-halopyridines were examined. While the enantioselectivity remained largely consistent, a decreasing trend in yield was observed in the order I > Br > Cl, which may correlate with increasing steric hindrance of the halogen substituents (Table 1, entries 8–10). Despite the reduced reactivity with bulkier halides, the reactions proceeded with moderate efficiency (60–73% yield), suggesting the broader applicability of this method to a range of azaarene substrates.

Table 1. Reaction Optimization[a]



entry	ligand[e]	silane	X	yield (%)[b]	*ee* (%)[c]
1	(*S*,*S*)-Ph-BPE	Ph_3SiH	F	76	94
2	(*R*)-Tol-BINAP	Ph_3SiH	F	31	64
3	(*R*)-SEGPHOS	Ph_3SiH	F	68	44
4	(*S*,*S*)-Ph-BPE	Ph_2MeSiH	F	77	94
5	(*S*,*S*)-Ph-BPE	$PhMe_2SiH$	F	75	93
6	(*S*,*S*)-Ph-BPE	$(EtO)_2MeSiH$	F	33	88
7[d]	(*S*,*S*)-Ph-BPE	Ph_2MeSiH	F	86	96
8[d]	(*S*,*S*)-Ph-BPE	Ph_2MeSiH	Cl	73	96
9[d]	(*S*,*S*)-Ph-BPE	Ph_2MeSiH	Br	70	96
10[d]	(*S*,*S*)-Ph-BPE	Ph_2MeSiH	I	60	96

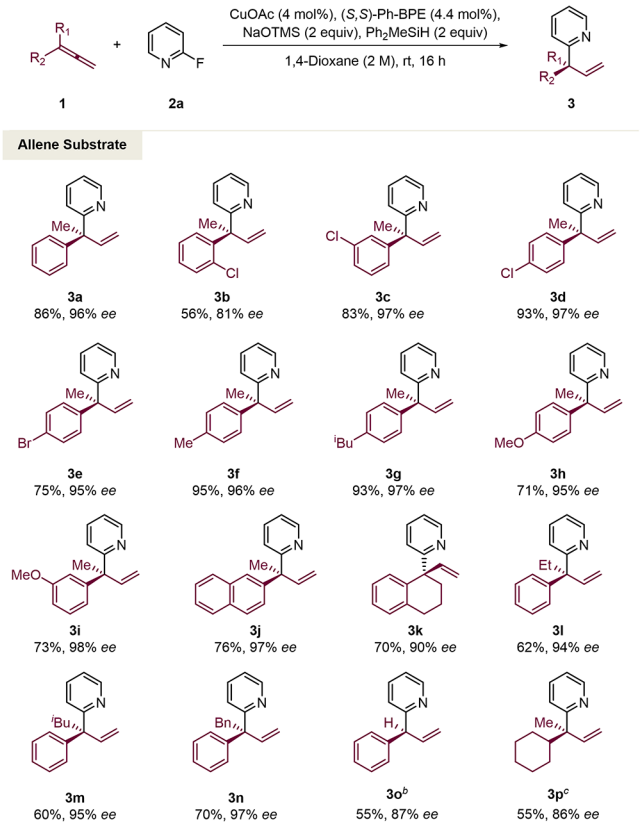
[a]Reactions were performed with **1a** (0.2 mmol), **2a** (0.6 mmol), silane (0.4 mmol), base (0.4 mmol), CuOAc (4 mol %), and ligand (4.4 mol %) in THF (2 M). [b]^{1}H NMR yields with mesitylene as the internal standard. [c]Enantiomeric excess was determined by HPLC. [d]1,4-Dioxane (2 M). [e]Ligand structures:



Next investigated the substrate scope of the reaction under the optimized conditions, beginning with a series of allene derivatives (Scheme 1). The position of the substituents on the aryl moiety of the allenes was found to exert a pronounced influence on the reactivity and enantioselectivity. When the substituent was located at the ortho position (**3b**), significant steric hindrance during the hydrocupration step led to diminished reactivity (56% yield, 81% *ee*), in contrast to the meta- (**3c**, 83% yield, 97% *ee*) and para-substituted analogues (**3d**, 93% yield, 97% *ee*). The reaction proceeded efficiently with a variety of functional groups, affording excellent yields (up to 95%) and enantioselectivities (up to 98% *ee*) for substrates bearing either electron-withdrawing substituents, such as halogens (**3b**–**3e**), or electron-donating groups,

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Scheme 1. Allene Substrate Scope[a]



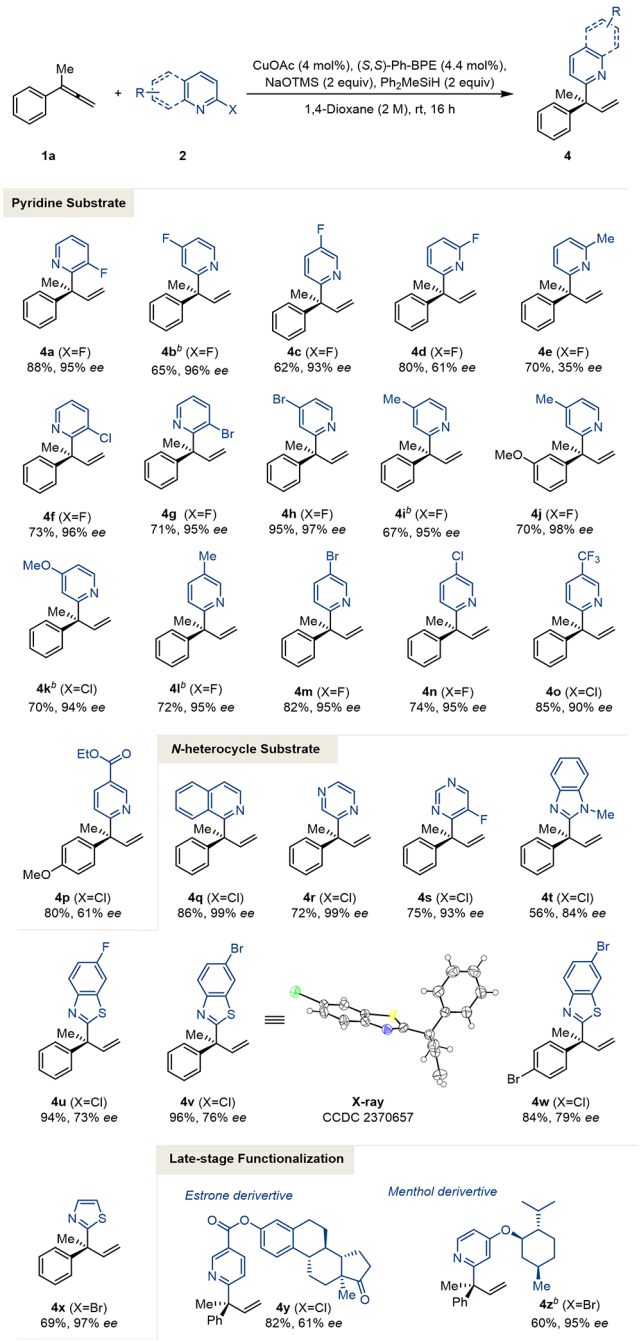
[a]Reaction conditions: **1** (0.2 mmol), **2a** (0.6 mmol), CuOAc (4 mol %), (*S*,*S*)-Ph-BPE (4.4 mol %), NaOTMS (0.4 mmol), and Ph_2MeSiH (0.4 mmol) in 1,4-dioxane (0.1 mL) at rt for 16 h. Isolated yields are shown. Enantiomeric excess was determined by HPLC. [b]The reaction was carried out at −10 °C for 24 h. [c]The reaction was carried out at 45 °C for 24 h.

including alkyl (**3f**, **3g**) and alkoxy groups (**3h**, **3i**). A naphthyl-substituted allene afforded the desired product **3j** in 76% yield with 97% *ee*. An exocyclic allene also proved to be compatible, delivering the product **3k** in 70% yield with 90% *ee*.

Extension of the alkyl chain length on the allene (**3l**−**3n**) resulted in slightly reduced yields (60−70%), while the enantioselectivity remained consistently high (94−97% *ee*). In the case of a monosubstituted allene, *β*-hydride elimination led to the formation of trisubstituted alkene byproducts. Performing the reaction at −10 °C suppressed this side reaction, enabling the formation of the desired product **3o** bearing a tertiary stereogenic center in moderate yield and enantioselectivity (55% yield, 87% *ee*; see Table S5). A dialkyl-substituted allene displayed low reactivity, likely due to its reduced nucleophilicity. Increasing the reaction temperature to 45 °C improved the conversion, affording the product **3p** in moderate yield (55%) and enantioselectivity (86% *ee*) (see Table S6).

We subsequently examined the scope of pyridine substrates amenable to the reaction conditions (Scheme 2). Substituents at the 3-, 4-, and 5-positions of the pyridine ring were well-tolerated, affording the corresponding products **4a**−**4c** in moderate to good yields (62−88%) with high enantioselectivities (93−96% *ee*). In contrast, 2,6-difluoropyridine delivered the monofunctionalized product **4d** in high yield (80%) but with markedly reduced enantioselectivity (61% *ee*). Moreover,

Scheme 2. *N*-Heterocycle Substrate Scope[a]

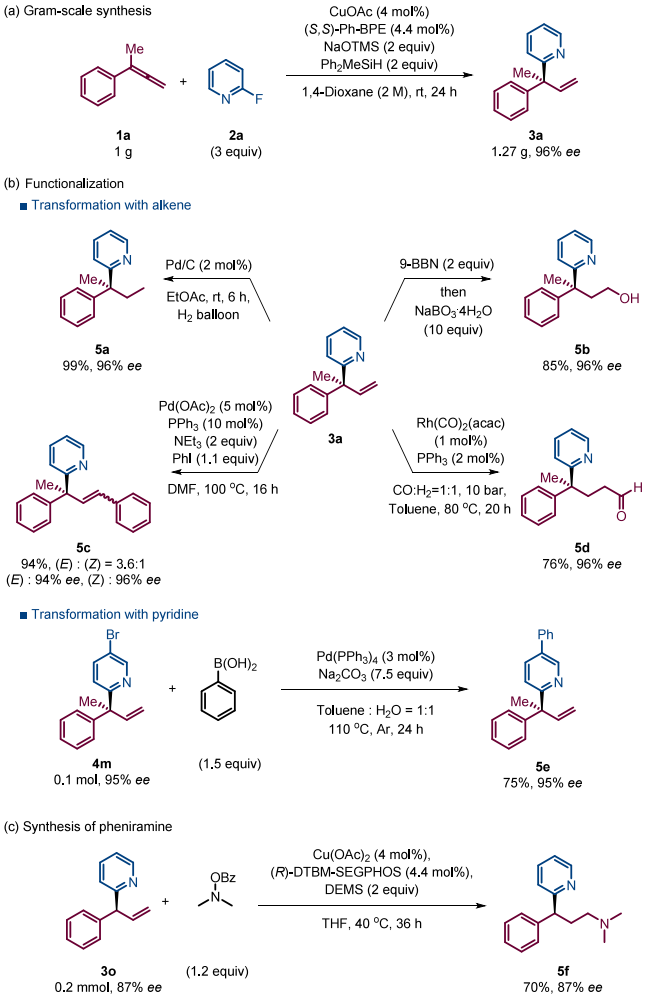


[a]Reaction conditions: **1a** (0.2 mmol), **2** (0.6 mmol), CuOAc (4 mol %), (*S*,*S*)-Ph-BPE (4.4 mol %), NaOTMS (0.4 mmol), and Ph_2MeSiH (0.4 mmol) in 1,4-dioxane (0.1 mL) at rt for 16 h. Isolated yields are shown. Enantiomeric excess was determined by HPLC. [b]Enantiomeric excess was determined by ^1H NMR with NMR shift reagents.

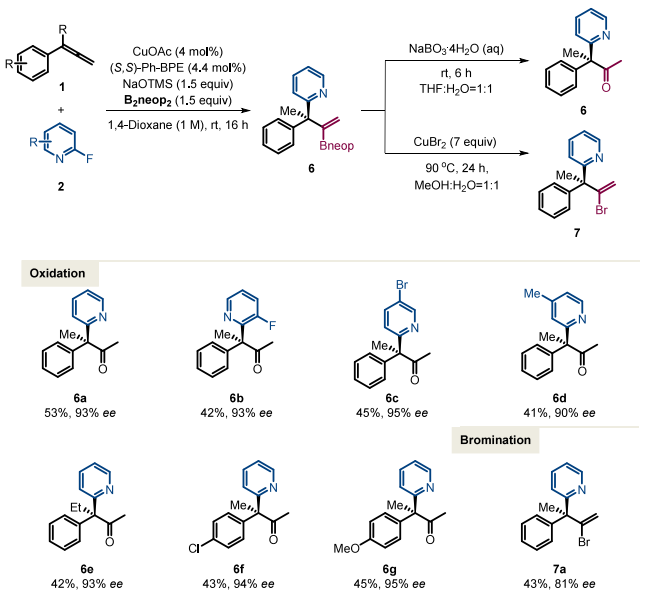
2-fluoro-6-methylpyridine gave the product **4e** in a similarly high yield (70%) but with even poorer enantioselectivity (35% *ee*). While substitution at the 3- or 5-position maintained a high *ee*, substitution at the sterically demanding 6-position consistently diminished it. This decrease became more pronounced as the size of the 6-substituent increased, suggesting that the steric congestion at this position interferes with the pyridine insertion step.[10] The reaction exhibited

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Scheme 3. Gram-Scale Synthesis and Application of Products

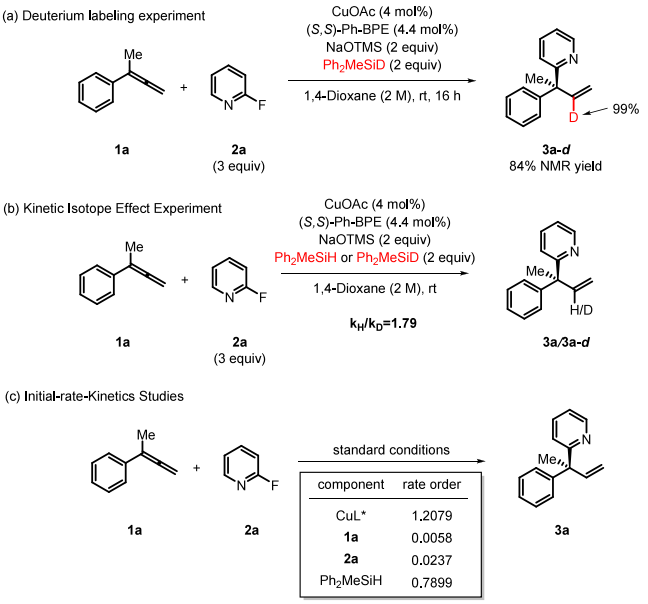


Scheme 4. Copper-Catalyzed Boropyridylation[a]



[a]Isolated yields are shown. Enantiomeric excess was determined by HPLC.

Scheme 5. Mechanistic Experiments



broad functional group tolerance, accommodating halogen, alkyl, and alkoxy substituents, which furnished the desired products (**4f**–**4k**) in up to 95% yield with up to 98% *ee*. A variety of 5-substituted pyridines—including those with methyl, halogens, and trifluoromethyl substituents—were also compatible, providing the corresponding products **4l**–**4o** in good yields (72–85%) with high enantioselectivities (90–95% *ee*). Although ethyl 6-chloronicotinate yielded the product **4p** in high yield (80%), it exhibited low enantioselectivity (61% *ee*), presumably because the ester group interfered with effective coordination between the pyridine nitrogen and the copper catalyst.

We further evaluated the applicability of this method to other nitrogen-containing heterocycles. Isoquinoline, pyrazine, and pyrimidine substrates all underwent the transformation smoothly, affording the corresponding products **4q**–**4s** in good yields (72–86%) with excellent enantioselectivities (93–99% *ee*). Benzimidazole and benzothiazole derivatives also participated effectively, delivering the desired products **4t**–**4w** in high yields (56–96%). However, these substrates exhibited slightly diminished enantioselectivities (73–84% *ee*), possibly due to increased steric hindrance around the reactive site. Notably, 2-bromothiazole furnished the product **4x** in good yield (69%) with excellent enantioselectivity (97% *ee*).

To further demonstrate the functional group compatibility of this methodology, the reaction was applied to pyridine substrates bearing complex natural-product-derived moieties, including estrone and menthol. In line with previous observations for ester-containing pyridines (e.g., **4p**), the estrone-derived substrate afforded the desired product **4y** in 82% yield but with diminished enantioselectivity (61% *ee*). In contrast, the menthol-derived substrate delivered the product **4z** in 60% yield with high enantioselectivity (95% *ee*).

We next assessed the scalability of the reaction under the optimized conditions (Scheme 3a). The desired product **3a** was successfully synthesized on a gram scale, affording 1.27 g of material with excellent enantioselectivity (96% *ee*). Given the synthetic versatility of the pyridine moiety bearing a terminal alkene, further functionalization of **3a** was explored (Scheme 3b). Hydrogenation and hydroboration–oxidation of

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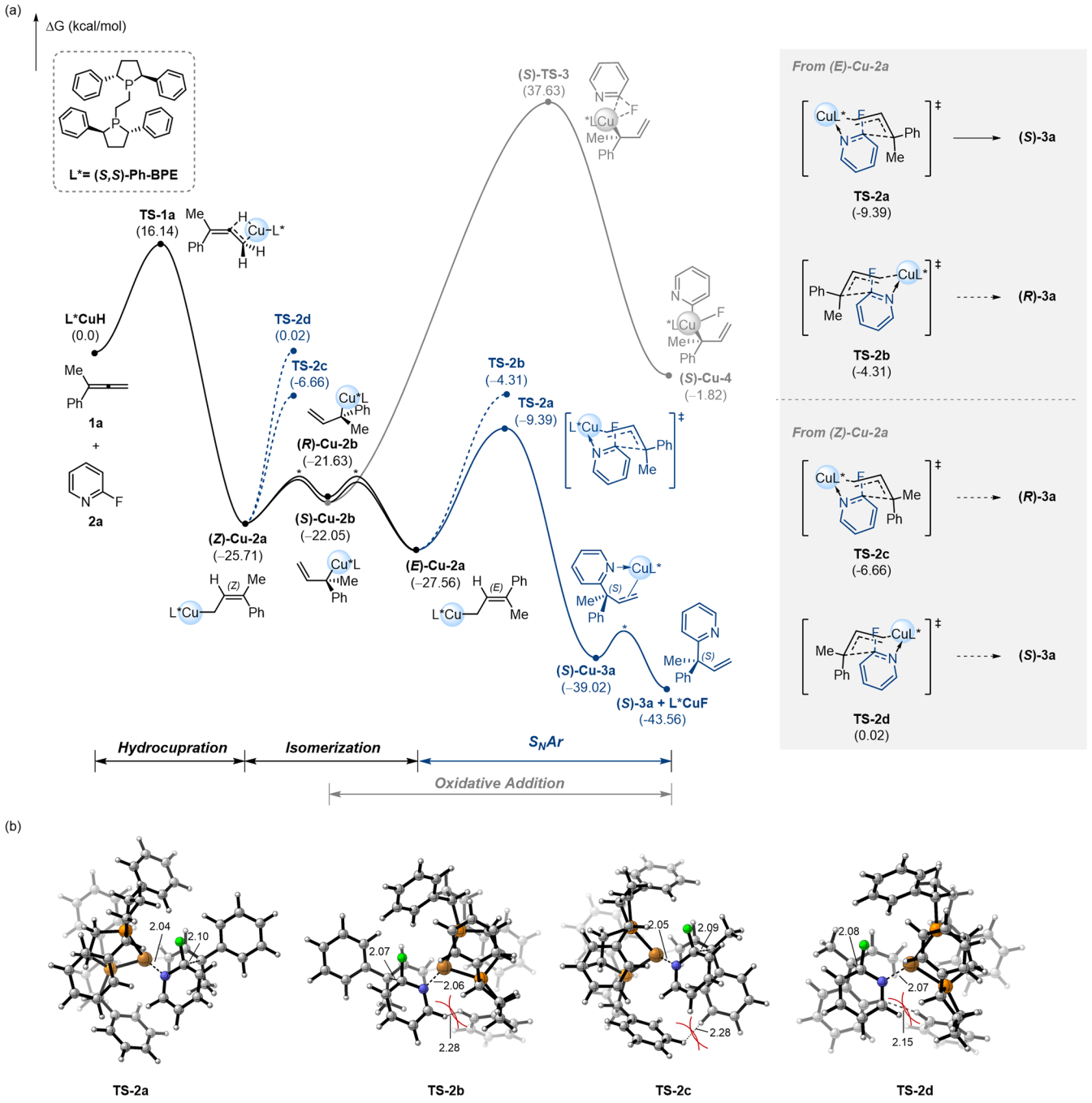


Figure 2. (a) Energy profile of the total reaction pathway. Transition states marked * were not located explicitly but were estimated. (b) Four transition states of the pyridine insertion step. All energies are in kcal/mol with respect to L*CuH, **1a**, and **2a**. DFT calculations were performed at the B3LYP-D3/SDD-6-311+G(d,p)/SMD(1,4-dioxane)//B3LYP-D3/LANL2DZ-6-31G(d,p) level of theory (see the Supporting Information for details).

the terminal alkene provided the corresponding saturated (**5a**) and alcohol (**5b**) derivatives, respectively, both in good yields with retention of enantioselectivity. Heck coupling furnished the internal alkene product **5c** with moderate stereoselectivity (*E*:*Z* = 3.6:1) while maintaining high enantioselectivity. Rhodium-catalyzed hydroformylation yielded the corresponding aldehyde **5d** in 76% yield with 96% *ee*. In addition, the halide functionality was exploited in a Suzuki cross-coupling of **4m**, affording the coupled product **5e** in 75% yield with 95% *ee*, with no erosion of enantioselectivity. Finally, hydroamination of **3o**, which bears both a tertiary stereogenic center and a terminal alkene, afforded the antihistamine drug pheniramine (**5f**) in 70% yield with moderate enantioselectivity (87% *ee*) (Scheme 3c).[14]

Our strategy was also applicable to the boropyridylation of allenes (Scheme 4). Under conditions analogous to those used for hydropyridylation—replacing Ph_2MeSiH with B_2neop_2—alkenylboron-containing products were obtained (see Tables S7–S9). In contrast, the use of B_2pin_2 or B_2cat_2 did not yield the desired products. CuOAc and 1,4-dioxane were identified as the optimal catalyst and solvent combination. Subsequent oxidation or bromination of the resulting boronates furnished

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ketone products (**6a**−**6g**) and an alkenyl bromide (**7a**), each bearing a quaternary stereogenic center, in moderate yields (41−53%) with good enantioselectivity (80−95% *ee*)

Next, we conducted a series of mechanistic studies to gain insight into the reaction mechanism (Scheme 5). When Ph_2MeSiD was used in place of Ph_2MeSiH, the deuterated product **3a-*d*** was obtained in 84% NMR yield with 99% deuterium incorporation (Scheme 5a). The deuterium was incorporated at C2 of the allene, consistent with a 1,2-hydrocupration pathway, as previously reported.[12,13] A kinetic isotope effect (KIE) experiment revealed a KIE value of 1.79, indicating that either the silane or copper hydride is involved in the rate-determining step (Scheme 5b). Further initial rate kinetic studies confirmed that both the copper and silane contribute to the rate-determining step, supporting that copper hydride generation is turnover-limiting (Scheme 5c).[10,15]

To investigate the origin of the enantioselectivity, we performed density functional theory (DFT) calculations. The proposed mechanism involves a sequence of hydrocupration, isomerization, and nucleophilic aromatic substitution (Figure 2a). Consistent with previously reported pathways,[12b,13] the reaction proceeds via a 2,3-hydrocupration pathway to generate intermediate (*Z*)-**Cu-2a** (see Figure S9 and S10). The transition state leading to (*Z*)-**Cu-2a** is at least 1 kcal/mol lower in energy than any other 1,2- or 2,3-hydrocupration transition state, indicating its kinetic favorability. Although halopyridines are known to engage in oxidative addition with transition metals,[8,16] our calculations revealed that the oxidative addition pathway involving 2-fluoropyridine has a prohibitively high activation barrier of 59.7 kcal/mol from the (*S*)-**Cu-2b** intermediate, effectively ruling out this pathway.

Following hydrocupration, 1,3-allylic rearrangement of (*Z*)-**Cu-2a** leads to the formation of multiple intermediates: (*Z*)-**Cu-2a**, (*E*)-**Cu-2a**, (*R*)-**Cu-2b**, and (*S*)-**Cu-2b**. Based on the observed stereochemical outcome of product **3a**, the S_NAr pathways involving (*R*)-**Cu-2b** and (*S*)-**Cu-2b** were excluded. When (*Z*)-**Cu-2a** and (*E*)-**Cu-2a** engage in S_NAr with 2-fluoropyridine, four possible transition states (**TS-2a**−**TS-2d**) can be considered, leading to either (*S*)-**3a** or (*R*)-**3a**. Among these, transition state **TS-2a**, derived from (*E*)-**Cu-2a**, was calculated to be the most favorable, with an energy of −9.4 kcal/mol. This is significantly lower than those of **TS-2b** (−4.3 kcal/mol), **TS-2c** (−6.7 kcal/mol), and **TS-2d** (0.0 kcal/mol). Accordingly, the (*E*)-**Cu-2a** intermediate undergoes S_NAr through **TS-2a** to furnish the product (*S*)-**3a** (Figure 2b). In **TS-2a**, steric repulsions between the ligand and substrate are minimized, and an additional stabilizing C−H···π interaction is observed,[17] contributing to its favorable energetics. In contrast, **TS-2b**, **TS-2c**, and **TS-2d** all suffer from significant steric repulsion, making them less favorable (Figure 2b).

CONCLUSION

In summary, we have developed a copper-catalyzed enantioselective strategy for the construction of all-carbon quaternary stereogenic centers adjacent to *N*-heterocycles via the hydropyridylation of 1,1-disubstituted allenes. This transformation proceeds under mild conditions with broad functional group tolerance and delivers a wide array of *N*-heterocycle-substituted products in high yields with excellent enantioselectivities (up to 99% *ee*). The methodology was further extended to copper-catalyzed boropyridylation, enabling access to synthetically versatile alkenylboronates that can be readily transformed into ketones, alcohols, or halogenated derivatives without erosion of the enantioselectivity. Gram-scale synthesis and derivatizations—including hydrogenation, hydroboration−oxidation, hydroformylation, and hydroamination—underscore the robustness and synthetic utility of this platform.

Mechanistic studies, including deuterium-labeling studies, kinetic isotope effect measurements, and initial rate kinetic experiments, indicate that copper hydride formation is turnover-limiting. DFT calculations support a reaction pathway involving regioselective 2,3-hydrocupration followed by a stereodefining nucleophilic aromatic substitution (S_NAr) step. The origin of enantioselectivity is attributed to favorable noncovalent interactions and minimized steric repulsion in the key transition state. Together, these findings establish a robust platform for the synthesis of highly congested azaarene architectures and open new avenues for asymmetric catalysis via nucleophilic aromatic substitution.

ASSOCIATED CONTENT

Supporting Information

The Supporting Information is available free of charge at https://pubs.acs.org/doi/10.1021/jacs.5c13377.

Detailed experimental procedures, spectral data, and analytical data (PDF)

Accession Codes

Deposition number 2370657 contains the supplementary crystallographic data for this paper. These data can be obtained free of charge via the joint Cambridge Crystallographic Data Centre (CCDC) and Fachinformationszentrum Karlsruhe Access Structures service.

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Notes

The authors declare no competing financial interest.

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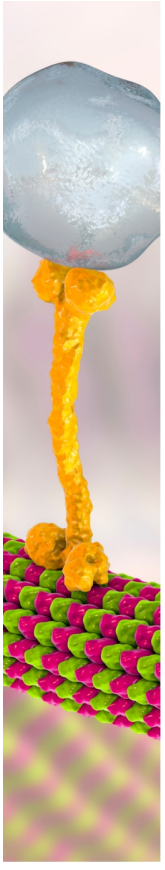
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